REALISING A PRESCRIBED FINITE RESIDUE FIELD EXTENSION BY A FINITE FREE RING MAP

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Induction on the degree of a finite field extension. Used at the start of the counted proof (lines 44902-44908). Reduces to the case where the extension is generated by a single element, by splitting off an intermediate field.
- The minimal polynomial of a primitive element. Used in the remaining case (lines 44910-44917). Supplies the monic relation whose coefficients are realised in R and then used to define S.

Lemma 0.1. *Let R be a ring. Let $\mathfrak{p} \subset R$ be a prime and let $L/\kappa(\mathfrak{p})$ be a finite extension of fields. Then there exists a finite free ring map $R \to S$ such that $\mathfrak{q} = \mathfrak{p}S$ is prime and $\kappa(\mathfrak{q})/\kappa(\mathfrak{p})$ is isomorphic to the given extension $L/\kappa(\mathfrak{p})$.*

Proof. By induction of the degree of $\kappa(\mathfrak{p}) \subset L$. If the degree is 1, then we take $R = S$. In general, if there exists a sub extension $\kappa(\mathfrak{p}) \subset L' \subset L$ then we win by induction on the degree (by first constructing $R \subset S'$ corresponding to $L'/\kappa(\mathfrak{p})$ and then construction $S' \subset S$ corresponding to L/L'). Thus we may assume that $L \supset \kappa(\mathfrak{p})$ is generated by a single element $\alpha \in L$. Let $X^d + \sum_{i<d} a_i X^i$ be the minimal polynomial of α over $\kappa(\mathfrak{p})$, so $a_i \in \kappa(\mathfrak{p})$. We may write a_i as the image of f_i/g for some $f_i, g \in R$ and $g \notin \mathfrak{p}$. After replacing α by $g\alpha$ (and correspondingly replacing a_i by $g^{d-i}a_i$) we may assume that a_i is the image of some $f_i \in R$. Then we simply take $S = R[x]/(x^d + \sum f_i x^i)$. □

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